Lazuli Science Memo: Cross-checking the `lazuli-exo` IFS Transmission-Spectroscopy Simulator Against As-Measured Data

G. Stefansson

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Document Version History

Version	Date	Authors	Change Description	Approved By / Date
0.1	2026-09-26	GS (drafted with Claude Code)	Initial draft; pilot checks A and B run	
0.2	2026-10-05	GS (drafted with Claude Code)	Checks A, B, C, D and G run on the regenerated (Vega/AB-corrected) tables and on the parametric path; package renamed `lazuli-exo`; JWST data added (WASP-39 b, TRAPPIST-1 e); references verified against arXiv; checks E, F, H not run. Corrected: WASP-107 $\Delta(V-J)$ from +0.27 to +0.09 (SIMBAD V now 11.47).	

Summary of decisions

`lazuli-exo` predicts Lazuli IFS transmission spectra and error bars for 14 showcase planets (PO-SEIDON tables) and for any planet (parametric model, `lazuli_exo.simulate`). Version 0.1 of this memo defined eight cross-checks against *as-measured* data and ran two pilots, one of which found the Vega/AB magnitude bug. That bug is fixed and the noise tables were regenerated on 2026-10-04. This version runs checks A–D and G against public archives and published JWST and HST data. All numbers below are derived here unless marked otherwise; scripts and data snapshots are in `memo/scripts/` and `memo/data/`.

- **The photon-normalization chain is now exactly right (check B(i)).** An independent hand budget $F_\lambda\,\lambda/(hc)\,A\,\eta\,\mathrm{QE}\,\Delta\lambda\,t_{\rm int}$, normalized to the catalog J with sncosmo, reproduces slicersim's extracted signal to 1.000 for all 14 hosts at every wavelength (Table B1). The Vega/AB fix is confirmed end to end.
- **Against JWST, Lazuli's predicted errors are consistent with collecting-area scaling (check B(ii)).** For WASP-39 b, lazuli-exo predicts a single-transit precision 2.3× worse than NIRISS/SOSS achieved in the same 0.004 μm bins, against 1.9× from area alone; for NIRSpec/PRISM it predicts 1.2× worse (Table B2). Against 23 of 27 HST datasets Lazuli predicts errors 0.07–0.66× the achieved values, well below the 0.75 area ratio: HST's duty cycle and systematics, not Lazuli's photons, set those. Lazuli's bars should therefore be read as a photon floor.
- **Stellar SEDs pass for hot hosts, not for M dwarfs (check A).** J-normalized model flux over Gaia DR3 XP flux is within 10 % at 0.40–1.02 μm for the eight hosts hotter than 4400 K (0.95–1.10 in the blue). The star tables over-predict the blue flux of GJ 1214 by 1.40× and GJ 3470 by 1.20×. The parametric path's PHOENIX mini-grid has corrupt nodes at non-solar metallicity below 3500 K: LHS 1140 comes out 0.63× XP in the blue (GitHub #16, new).
- **The registry is not self-consistent for two targets (check C).** t_{14} recomputed from the registry's own geometry is 0.85× the registry t_{14} for HAT-P-11 b and 0.93× for LHS 1140 b; all other targets agree within 3.5 %. Across published solutions, k differs by 1–4 %, i.e. white-light depths by 150–1900 ppm, more than Lazuli's single-transit error for every target. The transit solver matches batman to 0.07 ppm; the local-arc $z(t)$ is off by up to 69 ppm (WASP-12 b).
- **The clear H_2/He scenario over-predicts every measured spectrum (check D).** Fitting data $= c + b\times$model, the clear scenario gives $b \leq 0.69$ for all 27 space-based datasets with at least 10 points, and $b < 0$ for two of them (Tables D1–D2, Figure 2). The scenario set brackets the measured amplitude for 23 of 47 datasets on the table path and 31 of 47 on the parametric path (15 and 22 of the 35 space-based ones). TRAPPIST-1 e's four JWST visits are flat ($b_{\rm clear} \leq 0.08$); WASP-107 b is nearly flat ($b_{\rm clear} = 0.09$). The reduced χ^2 of the best toy scenario against JWST-quality data is 5–6.
- **Stellar contamination is measurable and already matters for TRAPPIST-1 e (check G).** The four JWST/PRISM visits scatter by 220 ppm visit to visit at 1.0–1.7 μm, of which 180 ppm is in excess of the per-visit errors, and their mean depths differ by 430 ppm. Lazuli's single-transit error there is 330 ppm per channel, so after about three transits the unmodelled stellar term dominates (GitHub #8).
- **Not run:** E (fitted limb darkening), F (H4RG lab ramps) and H (slicersim time-series injection). E needs fitted coefficients that are not tabulated in the archive; F and H need internal inputs.

- **Decisions.** (1) Present "clear" as an upper envelope and make "cloudy" or "high metallicity" the default showcase scenario for sub-Neptunes; drop the H_2/He scenarios for TRAPPIST-1 e and LHS 1140 b. (2) Fix the PHOENIX mini-grid (#16) and normalize the M-dwarf showcase stars to Gaia XP. (3) Freeze one solution per target (#7). (4) Add a stellar-contamination term bounded by check G before quoting any M-dwarf S/N (#8). (5) Keep the "statistical noise only" caveat; the HST comparison shows why.

Background

`lazuli-exo` (formerly `terminator`) is a numpy-only runtime over precomputed tables: POSEIDON isothermal equilibrium-chemistry H_2/He scenarios for 14 planets, slicersim single-transit noise tables for the `lazuli_cbe` configuration, exotic-ld limb-darkening tables, and, since 2026-10-04, a planet-independent photospheric-pressure grid and a PHOENIX stellar mini-grid that let `simulate()` model any planet with slicersim computing the noise live. Target parameters are frozen from the NASA Exoplanet Archive composite table.

Version 0.1 of this memo argued that every validation to date was internal and proposed eight as-measured checks. Its pilot photon budget found that 2MASS J (Vega) was normalized as AB inside slicersim, making every host 2.3× too bright. That was fixed in `lazuli_exo.slicer.depth_error` and the noise tables were regenerated against slicersim 1.0.0 (`0947b53`). The 2026-10-05 review (`report/2026-10-05_report.md`) then found that the two model paths disagree with each other by 5–75 % in feature amplitude for the same planet because of the 10-bar reference-radius convention of the tables (GitHub #3). The checks below therefore test both paths wherever a model spectrum enters.

Modeling Approach

Each check isolates one stage of the forward model and compares it with a measured quantity. Throughout: model products come from the committed tables and the committed grids, never regenerated ad hoc; all scripts run in the `py313` environment from the repository root; every comparison is reported per target; the reference data are snapshotted in `memo/data/` by `fetch_data.py` so the checks rerun offline.

A. Stellar SED vs Gaia DR3 XP and SIMBAD photometry (`checkA_sed.py`). Both model stars (the star table that fed the noise tables, and the parametric path's PHOENIX mini-grid at the registry $T_{\rm eff}$, $\log g$, [Fe/H]) are normalized to the catalog 2MASS J (Vega) exactly as slicersim normalizes them, then divided by the absolute XP flux (XP_SAMPLED, 336–1020 nm; De Angeli et al. 2023, Montegriffo et al. 2023) in 50 nm bins. Synthetic $V-J$, $J-H$ and $H-K$ are compared with SIMBAD. K2-18 and TRAPPIST-1 have no XP product.

B. Photon budget and achieved precision (`checkB_photon.py`). (i) For each host, $N_{\rm ADU}(\lambda) = F_\lambda\,\lambda/(hc)\,A\,\eta(\lambda)\,{\rm QE}(\lambda)\,\Delta\lambda\,t_{\rm int}\,g$ is computed by hand with F_λ normalized to J by sncosmo, $A = 7.07\,{\rm m}^2$ and η, QE, $\Delta\lambda$, $t_{\rm int}$ read from the instrument configuration, and compared channel by channel with slicersim's extracted point-source signal for the same readout (no LSF). (ii) Every HST and JWST dataset in the archive with per-bin errors inside 0.40–1.70 μm is compared with lazuli-exo's single-transit error for the same bins, next to the area-only expectation $\sqrt{A_{\rm other}/A_{\rm Lazuli}}$ ($A_{\rm JWST} = 25.4\,{\rm m}^2$, stated by JWST User Documentation; $A_{\rm HST} = 4.0\,{\rm m}^2$, 2.4 m aperture less the secondary obstruction, derived here).

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C. White-light geometry (`checkC_geometry.py`). Every archive `ps` solution with $(k, a/R_\star, i)$ is compared with the registry, with t_{14} recomputed from each solution's own geometry (Winn 2010 form, `lazuli_exo.orbit`). The numpy transit solver and the local-arc $z(t)$ are compared with batman (Kreidberg 2015) per target. Fitting TESS light curves, proposed in v0.1, was not done.

D. Spectral shape (`checkD_spectra.py`). Every archive dataset (any facility) with at least five points and errors inside 0.40–1.70 μm, plus the Espinoza et al. (2025) TRAPPIST-1 e JWST/PRISM and HST/WFC3 visits, is compared with all five scenarios on both model paths. Each scenario is binned to the dataset's bins and fitted two ways: data = c+ model (offset only, giving χ^2_ν) and data = $c + b\times$model (giving the data/model amplitude ratio b). A dataset is "bracketed" when the four atmospheric scenarios' b values straddle 1. The purpose is representativeness, not goodness of fit.

E. Limb darkening vs fitted coefficients. Not run; the archive does not tabulate fitted LD coefficients.

F. Detector model vs H4RG lab ramps. Not run; needs the detector group's data.

G. Stellar contamination from JWST epoch variability (`checkG_tlse.py`). The four TRAPPIST-1 e PRISM visits (Espinoza et al. 2025, Zenodo 16125662) give the visit-to-visit scatter of the depth at each wavelength; its excess over the per-visit errors bounds the transit-light-source term.

H. End-to-end injection and recovery. Not run; needs slicersim time series.

Results

A. Stellar SEDs

Host	$T_{\rm eff}$	table star	$\Delta(V-J)$	$\Delta(J-H)$	blue	red	$\Delta(V-J)$	blue	red
GJ 1214	3101	phoenix	-0.50	-0.03	1.40	1.04	-0.30	1.15	0.93
GJ 3470	3600	phoenix	–	0.07	1.20	1.05	–	0.78	0.94
HAT-P-11	4653	phoenix	–	0.13	0.97	1.00	–	0.95	0.98
HD 189733	5052	phoenix	0.01	0.01	1.01	0.99	0.01	1.01	0.99
HD 209458	6091	phoenix	-0.03	0.06	1.01	1.04	-0.03	1.03	1.04
K2-18	3457	phoenix	-0.22	0.02	–	–	-0.03	–	–
LHS 1140	3096	phoenix	-0.06	0.11	1.01	0.92	0.48	0.63	0.78
TOI-270	3506	phoenix	-0.10	0.08	1.03	0.97	-0.03	0.97	0.95
TRAPPIST-1	2566	phoenix	-0.45	-0.03	–	–	-0.48	–	–
WASP-107	4425	phoenix	0.09	0.07	1.10	1.00	0.09	1.11	1.00
WASP-12	6265	phoenix	-0.17	0.01	1.20	1.10	-0.19	1.23	1.11
WASP-121	6628	phoenix	-0.11	0.02	1.04	1.02	-0.15	1.09	1.04
WASP-39	5485	phoenix	-0.12	0.03	1.09	1.04	-0.12	1.09	1.04
WASP-96	5540	phoenix	0.34	0.05	0.99	1.00	0.34	0.98	1.01

Table A. Check A. Columns 4–7: the committed star table (POSEIDON PHOENIX, or blackbody); columns 8–10: the parametric path's PHOENIX mini-grid. Δ = synthetic minus SIMBAD colour (Vega). "blue" and "red" are the J-normalized model photon flux over the Gaia DR3 XP flux integrated over 0.40–0.70 and 0.70–1.02 μm (1.00 = perfect). K2-18 and TRAPPIST-1 have no XP spectrum; GJ 3470 and HAT-P-11 have no SIMBAD V. Derived here.

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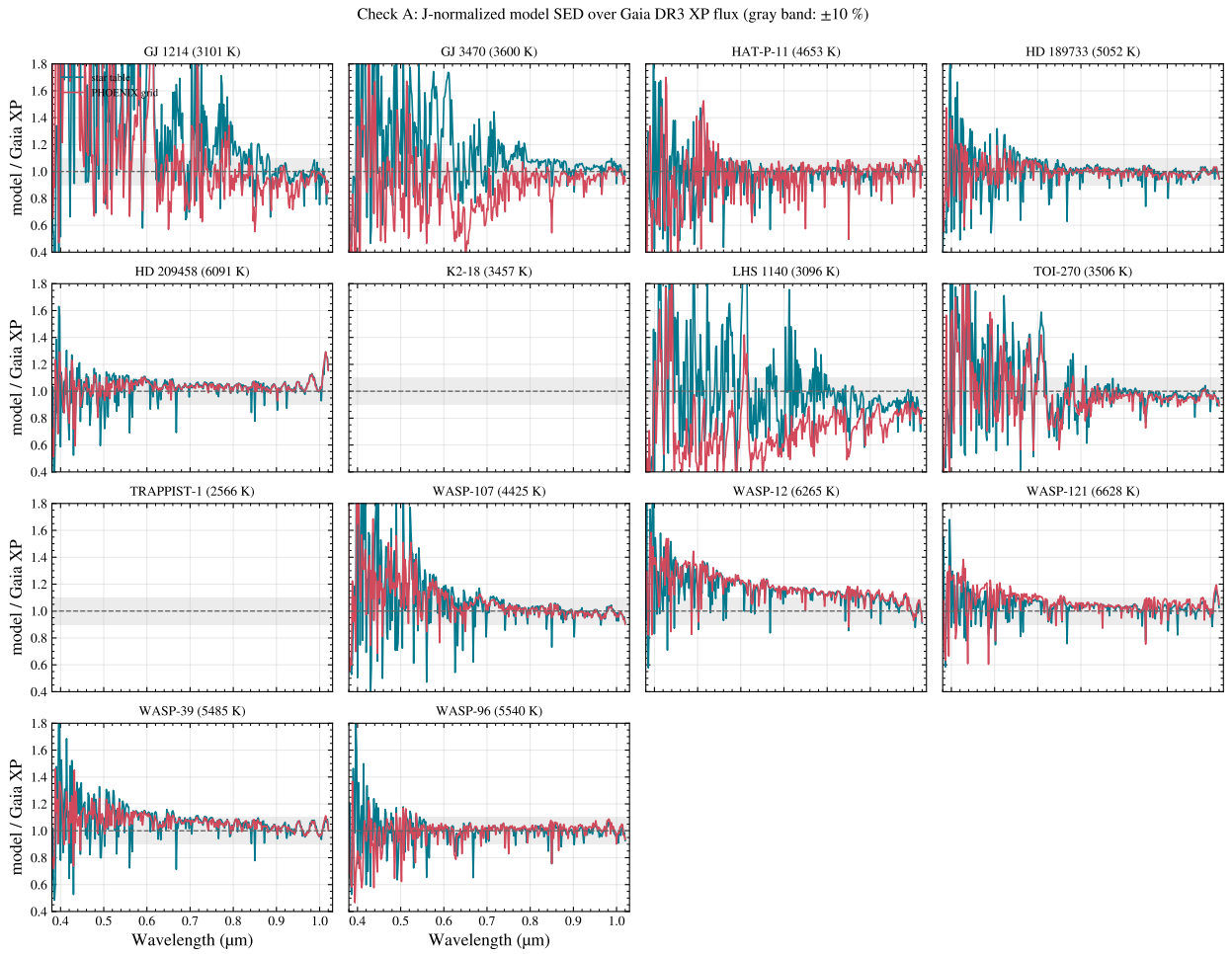


Figure 1. J-normalized model SED over Gaia DR3 XP flux for the twelve hosts with an XP spectrum. Teal: the committed star table; red: the parametric path's PHOENIX mini-grid. The gray band is ±10 %. The ratio is the error Lazuli's blue channels inherit from the stellar model. Derived here.

For the eight hosts hotter than 4400 K both models sit within the ±10 % band over 0.40–1.02 μm (blue ratios 0.95–1.10; WASP-12 is the worst at 1.20–1.23, a hot-star metallicity effect). The M dwarfs fail. The star table over-predicts GJ 1214's blue flux by 1.40× and GJ 3470's by 1.20×, consistent with the $V-J$ colours being 0.3–0.5 mag too blue for GJ 1214 and TRAPPIST-1 (PHOENIX runs hot in the optical for late-M dwarfs). The parametric path's mini-grid is worse in a different way: LHS 1140 is 0.63× and GJ 3470 0.78× XP in the blue. Evaluating the grid at its nodes shows the cause: at 3100 K the [Fe/H] $= -0.5$, $\log g = 5.0$ node and the [Fe/H] $= +0.3, +0.5$ nodes at $\log g \leq 4.5$ have synthetic $V-J$ of 5.9–6.7 against 4.4–4.7 for their neighbours. The low-resolution non-solar pysynphot PHOENIX folders carry placeholder-like columns that the all-zero test in `make_phoenix_grid.py` does not catch, and trilinear interpolation drags every cool, non-solar host toward them (GitHub #16). The pilot's WASP-96 anomaly ($\Delta(V-J) = +0.34$) is a catalog V problem: its XP ratio is 0.99.

B. Photon budget and achieved precision

Planet	J_{Vega}	J_{AB}	MACC	t_{int} (s)	slicersim / hand	16–84%	var / signal
GJ 1214 b	9.75	10.65	2:8:0	22.6	1.000	1.000–1.000	0.73
GJ 3470 b	8.79	9.69	2:4:0	11.3	1.000	1.000–1.000	0.73
HAT-P-11 b	7.61	8.51	2:1:0	2.8	1.000	1.000–1.000	1.07
HD 189733 b	6.07	6.97	2:1:0	2.8	1.000	1.000–1.000	1.17
HD 209458 b	6.59	7.49	2:1:0	2.8	1.000	1.000–1.000	1.05
K2-18 b	9.76	10.66	2:8:0	22.6	1.000	1.000–1.000	0.70
LHS 1140 b	9.61	10.51	2:8:0	22.6	1.000	1.000–1.000	0.72
TOI-270 d	9.10	10.00	2:4:0	11.3	1.000	1.000–1.000	0.74
TRAPPIST-1 e	11.35	12.25	8:8:0	158.5	1.000	1.000–1.000	1.22
WASP-107 b	9.38	10.28	2:8:0	22.6	1.000	1.000–1.000	0.68
WASP-12 b	10.48	11.38	2:8:0	22.6	1.000	1.000–1.000	0.69
WASP-121 b	9.62	10.52	2:8:0	22.6	1.000	1.000–1.000	0.68
WASP-39 b	10.66	11.56	4:8:0	67.9	1.000	1.000–1.000	0.93
WASP-96 b	11.27	12.17	4:8:0	67.9	1.000	1.000–1.000	0.93

Table B1. Check B(i). Extracted slicersim signal over the hand photon budget $F_\lambda\,\lambda/(hc)\,A\,\eta\,\mathrm{QE}\,\Delta\lambda\,t_{\mathrm{int}}$ (median over 0.40–1.70 μm, with the 16–84% range), where F_λ is the star table normalized here to the catalog J with sncosmo and $A = 7.07\,\mathrm{m}^2$, η and QE are read from the `lazuli_cbe` configuration. The last column is the Rauscher (2007) up-the-ramp variance over the Poisson variance of the accumulated signal. Derived here.

The hand budget and slicersim agree to three decimals for all 14 hosts and in all three wavelength thirds. This closes the pilot's finding: with the Vega-to-AB conversion in `slicer.depth_error`, the magnitude system, collecting area, throughput, quantum efficiency, channel width and integration time are all applied once and consistently. The variance-over-signal column is the Rauscher et al. (2007) up-the-ramp expression: below 1 for the frame-averaged (2,8,0) and (2,4,0) ramps, above 1 where read noise contributes ((2,1,0) ramps on the bright hosts; (8,8,0) on TRAPPIST-1).

Planet	Dataset	Fac.	n	bin (μm)	achieved σ	Lazuli σ_1	ratio	area-only
GJ 1214 b	Berta et al. 2012	HST	24	0.023	212	77	0.36↓	0.75
GJ 1214 b	Kreidberg et al. 2014	HST	22	0.023	28	76	2.74	0.75
GJ 3470 b	Benneke et al. 2019	HST	8	0.049	101	37	0.35↓	0.75
GJ 3470 b	Benneke et al. 2019	HST	18	0.030	32	28	0.92	0.75
GJ 3470 b	Ehrenreich et al. 2014	HST	107	0.005	398	73	0.18↓	0.75
HAT-P-11 b	Fraine et al. 2014	HST	29	0.019	43	24	0.59↓	0.75
HD 189733 b	McCullough et al. 2014	HST	28	0.019	120	19	0.15↓	0.75
HD 189733 b	Pont et al. 2013	HST	10	0.025	39	16	0.40↓	0.75
HD 189733 b	Pont et al. 2013	HST	6	0.027	246	14	0.07↓	0.75
HD 189733 b	Pont et al. 2013	HST	12	0.017	70	19	0.32↓	0.75
HD 189733 b	Sing et al. 2016	HST	18	0.025	41	16	0.38↓	0.75
HD 189733 b	Sing et al. 2016	HST	29	0.009	68	23	0.31↓	0.75
HD 209458 b	Deming et al. 2013	HST	28	0.019	35	15	0.40↓	0.75
HD 209458 b	Sing et al. 2016	HST	81	0.004	85	28	0.35↓	0.75
HD 209458 b	Sing et al. 2016	HST	29	0.009	34	21	0.60↓	0.75
K2-18 b	Benneke et al. 2019	HST	17	0.030	23	37	1.63	0.75
K2-18 b	Tsiaras et al. 2019	HST	17	0.028	24	39	1.63	0.75
WASP-107 b	Kreidberg et al. 2018	HST	20	0.026	57	34	0.59↓	0.75
WASP-107 b	Spake et al. 2018	HST	49	0.024	69	35	0.66↓	0.75
WASP-12 b	Kreidberg et al. 2015	HST	13	0.069	51	32	0.65↓	0.75
WASP-12 b	Mandell et al. 2013	HST	19	0.027	229	56	0.24↓	0.75
WASP-12 b	Sing et al. 2013	HST	20	0.052	125	37	0.29↓	0.75
WASP-12 b	Stevenson et al. 2014	HST	29	0.015	100	58	0.47↓	0.75
WASP-121 b	Evans et al. 2016	HST	31	0.019	87	45	0.50↓	0.75
WASP-39 b	Feinstein et al. 2022	JWST	213	0.004	60	136	2.30	1.90
WASP-39 b	Fischer et al. 2016	HST	31	0.010	263	99	0.36↓	0.75
WASP-39 b	Rustamkulov et al. 2022	JWST	63	0.006	85	117	1.17↓	1.90
WASP-39 b	Sing et al. 2016	HST	31	0.010	281	104	0.36↓	0.75
WASP-39 b	Wakeford et al. 2018	HST	36	0.019	172	66	0.41↓	0.75

Table B2. Check B(ii). Published per-bin depth precision inside 0.40–1.70 μm (median, ppm, as tabulated in the NASA Exoplanet Archive) against lazuli-exo's predicted single-transit precision for the same bins (committed noise tables; statistical noise only). "area-only" is $\sqrt{A_{\rm other}/A_{\rm Lazuli}}$ with $A_{\rm HST} = 4.0$, $A_{\rm JWST} = 25.4$, $A_{\rm Lazuli} = 7.07\,{\rm m}^2$; ↓ marks datasets where Lazuli's predicted error is smaller than that scaling. The archive does not record the number of visits combined, so ratios are indicative. Derived here.

The two JWST datasets are the clean comparison. For WASP-39 b, lazuli-exo predicts a single-transit precision of 136 ppm in the 0.004 μm NIRISS/SOSS bins where Feinstein et al. (2023) achieved 60 ppm, a ratio of 2.3 against 1.9 from collecting area alone; against the PRISM bins of Rustamkulov et al. (2023) the ratio is 1.2. Given that NIRISS and PRISM throughputs and their partial-saturation handling for a $J = 10.7$ star are not in this comparison, the predicted Lazuli precision is consistent with photon-limited scaling to within a factor 1.5. Against HST, Lazuli predicts errors 0.07–0.66× the achieved per-bin values for 23 of 27 datasets, far below the 0.75 area ratio. HST's ∼50 % orbital duty cycle, scanning overheads and systematics explain most of that, but it also means lazuli-exo's statistical-only bars are a floor, not a forecast. The exceptions (K2-18 b at 1.6 in both reductions, GJ 1214 b Kreidberg et al. at 2.7, GJ 3470 b Benneke et al. WFC3 at 0.9) are multi-transit programs compared here with one Lazuli transit. HD 189733 b and HD 209458 b sit on saturated noise tables and are flagged unvalidated in the code.

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C. White-light geometry

Planet	$t_{14}^{\rm geo}/t_{14}$	b	$n_{\rm ref}$	$\Delta k/k$	$\Delta\delta$ (ppm)	latest reference	$k/k_{\rm reg}$	$a/R_\star$ ratio	$t_{14}^{\rm geo}/t_{14}^{\rm reg}$
GJ 1214 b	1.000	0.26	8	0.031	850	Mahajan et al. 2024	1.000	1.000	1.000
GJ 3470 b	1.014	0.21	4	0.044	525	Kokori et al. 2023	1.000	0.994	1.008
HAT-P-11 b	0.847	0.24	10	0.024	162	Basilicata et al. 2024	1.017	0.900	1.107
HD 189733 b	1.001	0.66	6	0.040	1925	Kokori et al. 2023	1.000	1.009	0.995
HD 209458 b	0.999	0.51	3	0.001	34	Kokori et al. 2023	1.000	0.994	1.006
K2-18 b	0.966	0.60	0	–	–	–	–	–	–
LHS 1140 b	0.928	0.22	4	0.042	449	Kokori et al. 2023	1.000	1.000	0.976
TOI-270 d	1.007	0.23	3	0.027	141	Mikal-Evans et al. 202	1.075	1.001	1.009
TRAPPIST-1 e	0.995	0.19	1	0.000	0	Agol et al. 2021	1.000	1.000	1.000
WASP-107 b	1.008	0.12	4	0.011	464	Wu et al. 2026	1.000	1.000	1.008
WASP-12 b	1.023	0.45	4	0.017	472	Kokori et al. 2023	1.007	1.013	0.998
WASP-121 b	0.997	0.10	4	0.028	847	Kokori et al. 2023	1.016	0.992	1.010
WASP-39 b	0.987	0.53	3	0.023	951	Kokori et al. 2023	1.000	0.998	1.028
WASP-96 b	1.014	0.71	2	0.020	560	Kokori et al. 2023	0.980	1.000	1.010

Table C1. Check C, part 1. Column 2: t_{14} recomputed from the registry's own $(a/R_\star, i, e, \omega, k)$ over the registry t_{14}. Columns 4–6: number of archive solutions with $(k, a/R_\star, i)$, their spread in k and in white-light depth. Columns 7–10: the most recent such solution against the registry. Derived here.

Planet	solver (ppm)	local arc (ppm)	$\sigma_{\rm pt}$ (ppm)	$t_{14}^{\rm batman}/t_{14}^{\rm reg}$	LD $I<0$	max $u_1{+}u_2$
GJ 1214 b	0.01	0.7	72	0.998	0	0.938
GJ 3470 b	0.01	2.2	63	1.014	0	0.888
HAT-P-11 b	0.01	23.7	43	0.844	0	0.887
HD 189733 b	0.01	0.0	32	0.999	0	0.914
HD 209458 b	0.01	0.0	25	0.999	0	0.874
K2-18 b	0.01	18.6	70	0.961	0	0.899
LHS 1140 b	0.01	0.0	68	0.924	0	0.949
TOI-270 d	0.01	0.2	74	1.004	0	0.857
TRAPPIST-1 e	0.01	0.0	77	0.920	6	1.048
WASP-107 b	0.07	5.8	54	1.001	0	0.895
WASP-12 b	0.01	68.8	79	1.014	0	0.864
WASP-121 b	0.07	7.2	52	0.996	0	0.857
WASP-39 b	0.01	0.0	61	0.978	0	0.903
WASP-96 b	0.01	0.0	81	1.006	0	0.897

Table C2. Check C, part 2. Numpy transit solver vs batman at identical separations (max $|\Delta F|$); lazuli-exo's local-arc $z(t)$ vs batman's Keplerian $z(t)$ for the white-light curve; the per-sample noise at native cadence for one transit; batman's t_{14} from the registry geometry over the registry t_{14}; number of LD chunks with negative limb intensity. Derived here.

The registry's own $(a/R_\star, i, e, \omega, k)$ reproduce its t_{14} within 3.5 % for twelve targets but give 0.85× for HAT-P-11 b and 0.93× for LHS 1140 b, confirming the audit's finding with independent per-reference solutions. Published k values spread by 1–4 % per target, which is 150–1900 ppm in white-light depth, larger than Lazuli's single-transit white-light error (25–80 ppm, Table C2) for every target. The numpy solver agrees with batman to 0.07 ppm; the local circular-arc $z(t)$ is within 1 ppm for circular orbits and off by 19–69 ppm for the eccentric or grazing ones (K2-18 b, HAT-P-11 b, WASP-12 b). TRAPPIST-1 e's limb-darkening table has six blue chunks with negative limb intensity.

D. Spectral shape

Planet	Dataset	Fac.	n	amp	σ	best	χ^2_ν	$\chi^2_{\nu,\mathrm{flat}}$	b_{clear}	b_{cloudy}	b_{hazy}
GJ 1214 b	Bean et al. 2011, MMIRS	grou	36	1032	175	flat	1.3	1.3	-0.09	0.30	0.41
GJ 1214 b	Berta et al. 2012, WFC3	HST	24	778	212	cloudy	0.6	0.8	0.17	1.24	0.18
GJ 1214 b	Croll et al. 2011, WIRCam	grou	5	780	340	cloudy	0.9	0.9	0.26	1.49	0.28
GJ 1214 b	Kreidberg et al. 2014, Wide Field C	HST	22	99	28	flat	1.0	1.0	0.01	0.09	0.01
GJ 1214 b	Rackham et al. 2017, IMACS spectr	grou	14	619	185	flat	0.8	0.8	-0.16	-3.48	-0.41
GJ 1214 b	Wilson et al. 2014, OSIRIS	grou	8	2299	423	hazy	4.0	4.4	0.55	53.00	5.20
GJ 3470 b	Benneke et al. 2019, Space Telesc	HST	8	303	101	cloudy	1.2	1.2	0.13	0.96	-0.21
GJ 3470 b	Benneke et al. 2019, Wide Field C	HST	18	161	32	cloudy	2.3	2.3	0.08	0.50	0.09
GJ 3470 b	Biddle et al. 2014,	grou	8	1536	291	hazy	0.9	1.3	0.40	-95.75	2.28
GJ 3470 b	Biddle et al. 2014, SBIG imager	grou	5	596	402	hazy	0.3	0.8	3.81	-45.23	2.27
GJ 3470 b	Dragomir et al. 2015, SBIG imager	grou	7	1492	476	hazy	0.5	0.9	-0.74	99.69	2.65
GJ 3470 b	Ehrenreich et al. 2014, Wide Field C	HST	107	2598	398	cloudy	1.4	1.4	0.20	0.81	0.20
HAT-P-11 b	Fraine et al. 2014, Wide Field C	HST	29	247	43	high-met	1.0	2.0	0.50	3.24	0.53
HD 189733 b	Barnes et al. 2016, HARPS Echell	grou	5	4500	1100	clear	2.6	2.6	–	–	–
HD 189733 b	Borsa et al. 2016, HARPS Echell	grou	7	676	149	hazy	0.5	1.9	-0.86	0.51	2.10
HD 189733 b	Di Gloria et al. 2015, HARPS Echell	grou	6	1152	365	hazy	0.9	1.1	-2.43	-11.40	2.68
HD 189733 b	McCullough et al. 2014, Wide Field C	HST	28	352	120	high-met	0.4	0.6	0.44	2.50	0.46
HD 189733 b	Pont et al. 2013, Advanced Cam	HST	10	461	39	hazy	1.2	6.5	0.63	4.73	0.78
HD 189733 b	Pont et al. 2013, Near Infrare	HST	6	1051	246	flat	2.8	2.8	-2.15	-15.42	-2.17
HD 189733 b	Pont et al. 2013, Space Telesc	HST	12	401	70	hazy	2.7	4.5	0.03	0.48	0.67
HD 189733 b	Sing et al. 2016, Space Telesc	HST	18	643	41	hazy	2.3	18.8	0.48	1.48	0.95
HD 189733 b	Sing et al. 2016, Wide Field C	HST	29	353	68	high-met	1.1	1.9	0.43	2.46	0.46
HD 209458 b	Deming et al. 2013, Wide Field C	HST	28	276	35	cloudy	1.6	5.9	0.32	1.07	0.33
HD 209458 b	Sing et al. 2016, Space Telesc	HST	81	701	85	cloudy	1.9	2.0	0.32	1.58	0.17
HD 209458 b	Sing et al. 2016, Wide Field C	HST	29	262	34	cloudy	2.6	5.1	0.26	0.79	0.27
K2-18 b	Benneke et al. 2019, Wide Field C	HST	17	102	23	high-met	1.2	1.9	0.60	5.16	0.63
K2-18 b	Tsiaras et al. 2019, Wide Field C	HST	17	115	24	high-met	1.2	1.6	0.44	4.32	0.45
WASP-107 b	Kreidberg et al. 2018, Wide Field C	HST	20	433	57	flat	6.5	6.5	0.09	0.39	0.10
WASP-107 b	Spake et al. 2018, Wide Field C	HST	49	623	69	flat	6.1	6.1	0.07	0.41	0.09
WASP-12 b	Kreidberg et al. 2015, Wide Field C	HST	13	273	51	cloudy	2.8	4.0	0.23	4.02	0.30
WASP-12 b	Mandell et al. 2013, Wide Field C	HST	19	646	229	high-met	0.4	0.6	0.69	5.21	0.70
WASP-12 b	Sing et al. 2013, Space Telesc	HST	20	569	125	flat	1.8	1.8	-0.34	-3.56	0.22
WASP-12 b	Stevenson et al. 2014, Wide Field C	HST	29	1170	100	flat	7.6	7.6	-0.59	-9.11	-0.82
WASP-121 b	Evans et al. 2016, Wide Field C	HST	31	1388	87	cloudy	4.0	4.2	0.08	0.75	0.32
WASP-39 b	Feinstein et al. 2022, Near Infrare	JWST	213	982	60	cloudy	5.4	13.1	0.22	0.82	0.23
WASP-39 b	Fischer et al. 2016, Space Telesc	HST	31	2067	263	cloudy	1.0	1.1	0.30	1.20	0.11
WASP-39 b	Nikolov et al. 2016, FORS2	grou	28	2017	327	cloudy	0.7	1.2	0.38	0.84	0.19
WASP-39 b	Rustamkulov et al. 202, Near Infrare	JWST	63	1337	85	cloudy	6.2	8.3	0.29	0.69	0.24
WASP-39 b	Sing et al. 2016, Space Telesc	HST	31	2122	281	cloudy	1.8	1.9	0.27	1.69	0.32
WASP-39 b	Wakeford et al. 2018, Wide Field C	HST	36	1128	172	cloudy	2.7	6.0	0.33	1.22	0.40
WASP-96 b	Nikolov et al. 2018, FORS2	grou	48	1135	221	clear	1.0	1.5	0.78	1.82	0.29
TRAPPIST-1 e	Espinoza et al. 2025, , HST_WFC3IR_v	HST	12	600	150	flat	1.4	1.4	-0.07	-0.00	-0.08
TRAPPIST-1 e	Espinoza et al. 2025, , HST_WFC3IR_v	HST	12	690	130	flat	2.2	2.2	-0.24	-1.65	-0.24
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	2045	152	flat	1.2	1.2	0.02	-0.09	0.00
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	6099	167	flat	0.7	0.7	0.08	0.46	0.07
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	2115	129	flat	1.0	1.0	0.01	-0.01	0.04
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	3343	214	flat	0.8	0.8	-0.02	-0.18	-0.06

Table D1. Committed POSEIDON tables. Check D. Every dataset with ≥ 5 points and errors inside 0.40–1.70 μm (NASA Exoplanet Archive; TRAPPIST-1 e from Espinoza et al. 2025). "amp" and σ: measured peak-to-peak depth and median error (ppm). "best" is the scenario with the lowest offset-fit χ^2_ν (also given for the flat scenario); b is the fitted data/model amplitude ratio (1 = the scenario has the measured amplitude); "brk." = the four atmospheric scenarios' b values straddle 1. Derived here.

Planet	Dataset	Fac.	n	amp	σ	best	χ^2_ν	$\chi^2_{\nu,\mathrm{flat}}$	b_{clear}	b_{cloudy}	b_{hazy}
GJ 1214 b	Bean et al. 2011, MMIRS	grou	36	1032	175	flat	1.3	1.3	-0.07	-0.00	0.44
GJ 1214 b	Berta et al. 2012, WFC3	HST	24	778	212	cloudy	0.6	0.8	0.22	2.32	0.24
GJ 1214 b	Croll et al. 2011, WIRCam	grou	5	780	340	high-met	0.8	0.9	0.33	2.66	0.36
GJ 1214 b	Kreidberg et al. 2014, Wide Field C	HST	22	99	28	flat	1.0	1.0	0.02	0.16	0.02
GJ 1214 b	Rackham et al. 2017, IMACS spectr	grou	14	619	185	flat	0.8	0.8	-0.23	-7.57	-0.53
GJ 1214 b	Wilson et al. 2014, OSIRIS	grou	8	2299	423	clear	4.0	4.4	1.20	119.41	4.72
GJ 3470 b	Benneke et al. 2019, Space Telesc	HST	8	303	101	cloudy	1.2	1.2	0.17	1.86	-0.20
GJ 3470 b	Benneke et al. 2019, Wide Field C	HST	18	161	32	cloudy	1.6	2.3	0.11	0.95	0.12
GJ 3470 b	Biddle et al. 2014,	grou	8	1536	291	hazy	1.0	1.3	3.81	-140.76	3.30
GJ 3470 b	Biddle et al. 2014, SBIG imager	grou	5	596	402	hazy	0.4	0.8	4.19	-38.78	3.11
GJ 3470 b	Dragomir et al. 2015, SBIG imager	grou	7	1492	476	hazy	0.7	0.9	-1.16	229.20	4.23
GJ 3470 b	Ehrenreich et al. 2014, Wide Field C	HST	107	2598	398	cloudy	1.4	1.4	0.26	1.40	0.27
HAT-P-11 b	Fraine et al. 2014, Wide Field C	HST	29	247	43	high-met	0.9	2.0	0.58	5.18	0.62
HD 189733 b	Barnes et al. 2016, HARPS Echell	grou	5	4500	1100	clear	2.6	2.6	–	–	–
HD 189733 b	Borsa et al. 2016, HARPS Echell	grou	7	676	149	hazy	0.8	1.9	-0.90	1.32	2.63
HD 189733 b	Di Gloria et al. 2015, HARPS Echell	grou	6	1152	365	hazy	0.9	1.1	-2.56	-13.45	3.03
HD 189733 b	McCullough et al. 2014, Wide Field C	HST	28	352	120	high-met	0.3	0.6	0.46	3.12	0.48
HD 189733 b	Pont et al. 2013, Advanced Cam	HST	10	461	39	hazy	1.3	6.5	0.66	6.07	0.79
HD 189733 b	Pont et al. 2013, Near Infrare	HST	6	1051	246	flat	2.8	2.8	-2.26	-20.39	-2.29
HD 189733 b	Pont et al. 2013, Space Telesc	HST	12	401	70	hazy	4.0	4.5	0.03	0.59	0.54
HD 189733 b	Sing et al. 2016, Space Telesc	HST	18	643	41	hazy	3.6	18.8	0.50	1.79	0.97
HD 189733 b	Sing et al. 2016, Wide Field C	HST	29	353	68	high-met	1.1	1.9	0.46	3.07	0.48
HD 209458 b	Deming et al. 2013, Wide Field C	HST	28	276	35	cloudy	2.1	5.9	0.35	1.48	0.37
HD 209458 b	Sing et al. 2016, Space Telesc	HST	81	701	85	cloudy	1.9	2.0	0.36	2.32	0.25
HD 209458 b	Sing et al. 2016, Wide Field C	HST	29	262	34	cloudy	2.5	5.1	0.29	1.07	0.30
K2-18 b	Benneke et al. 2019, Wide Field C	HST	17	102	23	high-met	1.2	1.9	0.69	7.91	0.73
K2-18 b	Tsiaras et al. 2019, Wide Field C	HST	17	115	24	high-met	1.1	1.6	0.50	6.73	0.53
WASP-107 b	Kreidberg et al. 2018, Wide Field C	HST	20	433	57	cloudy	2.0	6.5	0.12	0.76	0.14
WASP-107 b	Spake et al. 2018, Wide Field C	HST	49	623	69	cloudy	3.5	6.1	0.09	0.80	0.12
WASP-12 b	Kreidberg et al. 2015, Wide Field C	HST	13	273	51	cloudy	3.2	4.0	0.26	6.10	0.33
WASP-12 b	Mandell et al. 2013, Wide Field C	HST	19	646	229	hazy	0.4	0.6	0.79	8.96	0.82
WASP-12 b	Sing et al. 2013, Space Telesc	HST	20	569	125	flat	1.8	1.8	-0.39	-6.52	0.22
WASP-12 b	Stevenson et al. 2014, Wide Field C	HST	29	1170	100	flat	7.6	7.6	-0.67	-14.39	-0.88
WASP-121 b	Evans et al. 2016, Wide Field C	HST	31	1388	87	cloudy	4.0	4.2	0.10	1.22	0.35
WASP-39 b	Feinstein et al. 2022, Near Infrare	JWST	213	982	60	cloudy	5.9	13.1	0.28	1.38	0.30
WASP-39 b	Fischer et al. 2016, Space Telesc	HST	31	2067	263	cloudy	1.0	1.1	0.37	2.09	0.20
WASP-39 b	Nikolov et al. 2016, FORS2	grou	28	2017	327	cloudy	0.7	1.2	0.47	1.31	0.39
WASP-39 b	Rustamkulov et al. 202, Near Infrare	JWST	63	1337	85	cloudy	5.9	8.3	0.37	1.10	0.35
WASP-39 b	Sing et al. 2016, Space Telesc	HST	31	2122	281	cloudy	1.8	1.9	0.33	2.95	0.50
WASP-39 b	Wakeford et al. 2018, Wide Field C	HST	36	1128	172	cloudy	3.6	6.0	0.42	2.05	0.52
WASP-96 b	Nikolov et al. 2018, FORS2	grou	48	1135	221	clear	1.0	1.5	0.89	2.45	0.51
TRAPPIST-1 e	Espinoza et al. 2025, , HST_WFC3IR_v	HST	12	600	150	flat	1.4	1.4	-0.12	0.35	-0.15
TRAPPIST-1 e	Espinoza et al. 2025, , HST_WFC3IR_v	HST	12	690	130	flat	2.2	2.2	-0.46	-6.58	-0.51
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	2045	152	flat	1.2	1.2	0.04	-0.53	0.02
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	6099	167	cloudy	0.7	0.7	0.15	1.76	0.16
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	2115	129	flat	1.0	1.0	0.01	-0.09	0.08
TRAPPIST-1 e	Espinoza et al. 2025, , JWST_PRISM_E	JWST	32	3343	214	flat	0.8	0.8	-0.04	-0.74	-0.12

Table D2. Parametric model with the same showcase presets. Check D. Every dataset with ≥ 5 points and errors inside 0.40–1.70 μm (NASA Exoplanet Archive; TRAPPIST-1 e from Espinoza et al. 2025). "amp" and σ: measured peak-to-peak depth and median error (ppm). "best" is the scenario with the lowest offset-fit χ^2_ν (also given for the flat scenario); b is the fitted data/model amplitude ratio (1 = the scenario has the measured amplitude); "brk." = the four atmospheric scenarios' b values straddle 1. Derived here.

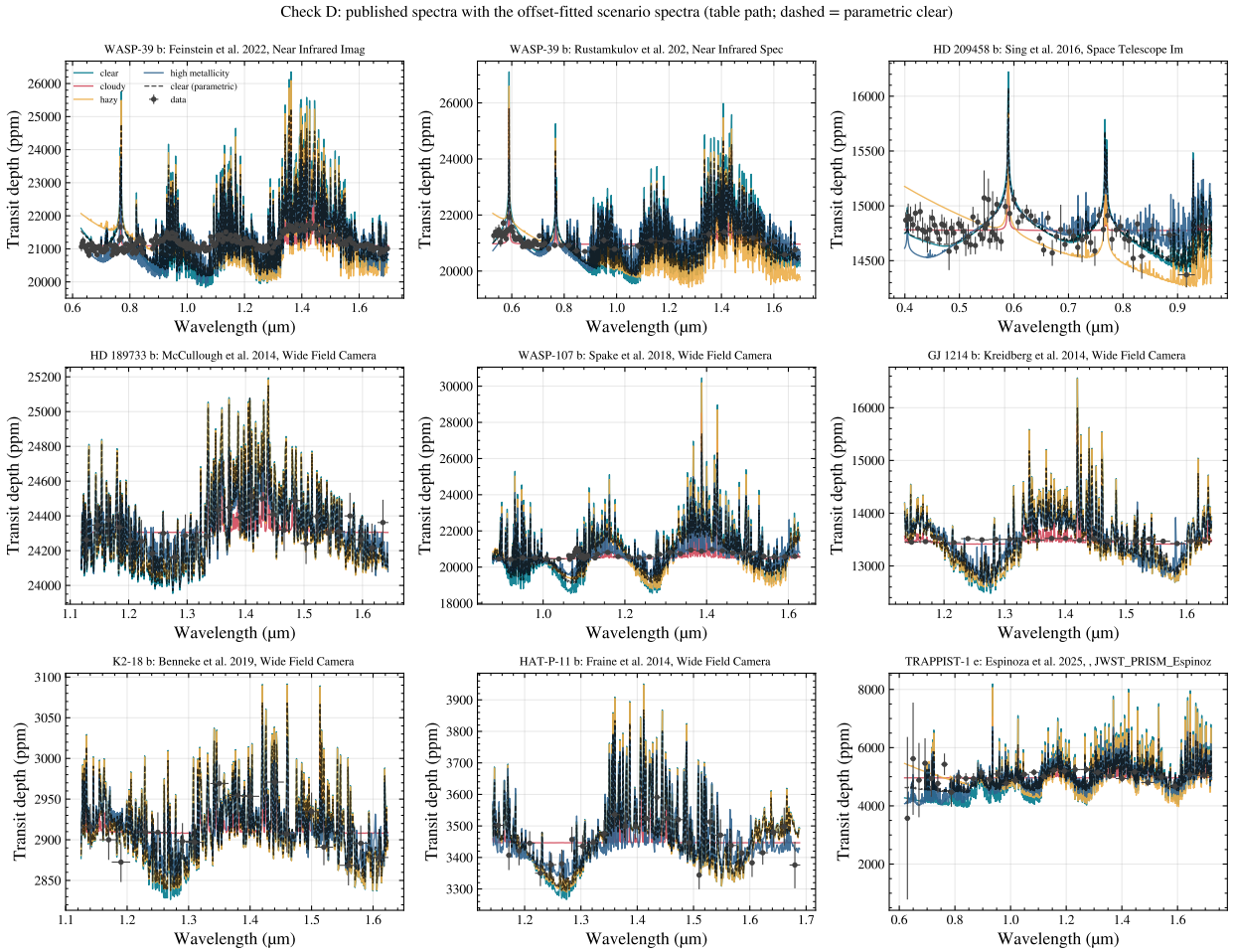


Figure 2. Published spectra (points, with bin widths) and the four atmospheric scenarios from the POSEIDON tables after an offset fit; the dashed line is the parametric clear scenario. Nine representative datasets, including the JWST NIRISS/SOSS and NIRSpec/PRISM spectra of WASP-39 b and the first JWST/PRISM visit of TRAPPIST-1 e. Derived here; data from the NASA Exoplanet Archive and Espinoza et al. (2025).

The clear H_2/He scenario over-predicts the measured feature amplitude everywhere: $b_{\rm clear} \leq 0.69$ for all 27 space-based datasets with at least ten points, and negative for the two WASP-12 b datasets whose structure anti-correlates with the model. The cloudy (1 mbar) scenario under-predicts most of them, so the set brackets the data for 23 of 47 datasets on the table path and 31 of 47 on the parametric path (15 and 22 of 35 space-based); the parametric path brackets more because its amplitudes are 10–50 % smaller (the reference-radius convention, GitHub #3), not because it is a better model. The best-fitting scenario is physically sensible where the data constrain it: high metallicity for K2-18 b (both WFC3 reductions), HAT-P-11 b and HD 189733 b WFC3; cloudy for HD 209458 b and WASP-39 b; flat for WASP-107 b and TRAPPIST-1 e. Against JWST-quality data no toy scenario fits ($\chi^2_\nu = 5$–6 for WASP-39 b), as expected for isothermal, gray-cloud models. TRAPPIST-1 e's four JWST visits and two HST visits are all best fitted by the flat scenario with $b_{\rm clear} \leq 0.08$: the H_2/He showcase scenarios for this planet are excluded by data, as the published analyses conclude (Espinoza et al. 2025). Ground-based datasets with 5–8 points are listed for completeness only; their b values are unconstrained.

G. Stellar contamination

band (μm)	n_λ	$\sigma_{\rm visit}$	std	excess	excess$_{84}$	offsets	Lazuli σ_1	clear	cloudy
0.6-1.0	15	309	287	0	352	431	1715	2280	749
1.0-1.7	17	126	221	180	237	431	327	3009	1306
0.6-1.7	32	160	243	168	259	431	396	3021	1306

Table G. Check G, TRAPPIST-1 e (Espinoza et al. 2025, four JWST/NIRSpec PRISM visits, 'Espinoza' reduction). All in ppm: median per-visit error; visit-to-visit standard deviation of the depth at each wavelength; the part of it in excess of the per-visit errors (median and 84th percentile); spread of the four visits' weighted-mean depths; lazuli-exo's single-transit error per native channel; and the parametric clear and cloudy H_2/He feature amplitudes over the same band. Derived here.

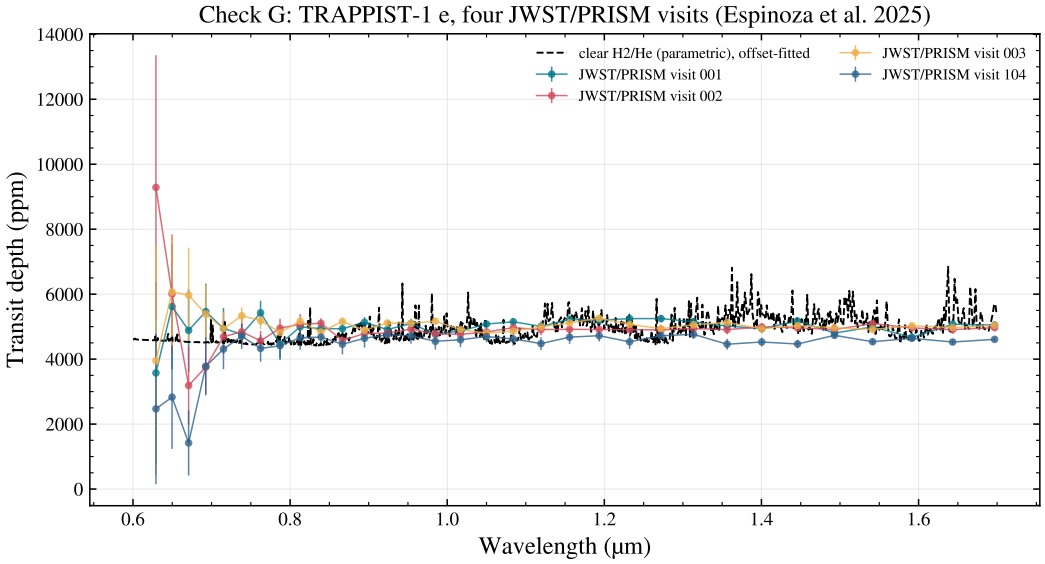


Figure 3. The four JWST/NIRSpec PRISM transmission spectra of TRAPPIST-1 e from Espinoza et al. (2025), 'Espinoza' reduction, over 0.6–1.7 μm, with the parametric clear H_2/He scenario after an offset fit. Derived here from the Zenodo record 16125662 data.

At 1.0–1.7 μm the four visits scatter by 220 ppm at fixed wavelength against a median per-visit error of 126 ppm, leaving 180 ppm of excess scatter (84th percentile 240 ppm); the visits' mean depths differ by 430 ppm. Espinoza et al. (2025) attribute this to the transit light source effect and note that current stellar models cannot reproduce it; Lim et al. (2023) find for TRAPPIST-1 b that the stellar-model uncertainty is an order of magnitude above their 89 ppm precision; Cadieux et al. (2024) detect unocculted faculae covering about 20 % of LHS 1140 at 5.8σ. Lazuli's single-transit error for TRAPPIST-1 e is 330 ppm per native channel at 1.0–1.7 μm, so after about three transits the unmodelled stellar term exceeds the statistical one, and it is already larger than the parametric cloudy-scenario amplitude in the blue.

Acceptance criteria

Check	Measured reference	Pass criterion (v0.1)	Status (v0.2)
A	Gaia XP, 2MASS, V	colours within 0.10 mag; XP within 10 % per 50 nm	pass $T_{\rm eff} > 4400$ K; fail M dwarfs
B(i)	hand photon budget	signal ratio 1.00 ± 0.05	pass (1.000)
B(ii)	achieved HST/JWST precision	$\sigma_{\rm Lazuli} \geq$ scaled photon floor	consistent with JWST; HST indicative
C	archive solutions, batman	t_{14} within 2 %; residual $< 0.2\sigma_{\rm pt}$	fail HAT-P-11 b, LHS 1140 b; solver pass; arc fail WASP-12 b
D	published spectra	scenario set brackets the data	23/47 (tables), 31/47 (parametric); clear always too large
E	fitted LD	$\lvert\Delta I(\mu)\rvert < 0.02$	not run
F	H4RG lab ramps	saturation onset within 5 %	not run
G	JWST M-dwarf epochs	TLSE bounded before a scenario is added	bounded: 180–430 ppm for TRAPPIST-1 e
H	slicersim time series	recovered scatter / table $= 1.00 \pm 0.05$	not run

Table 3. Acceptance criteria from v0.1 and the v0.2 outcome.

Discussion

Limitations.

- Check D cannot validate the toy scenarios; it shows which side of the data they fall on. It does show, with 27 space-based datasets, that "clear" is an envelope and never a central estimate.
- Check B(ii) inherits each reduction's systematics and the archive does not record how many transits a published spectrum combines. The JWST ratios are the ones to quote; the HST ratios say more about HST than about Lazuli.
- The hand budget in B(i) takes A, η and QE from the same configuration slicersim uses, so it validates units and normalization, not the throughput curve itself. Check F or a measured end-to-end throughput is needed for that.
- Gaia XP stops at 1.02 μm; the $J-H$ colours (within 0.13 mag for all hosts except GJ 3470 and LHS 1140 on the mini-grid, see Table A) are the only test of the red half of the band.
- Check G uses one reduction of one planet. The TRAPPIST-1 contamination level is extreme; the bound should be redone per host class before it becomes a scenario parameter.
- The Lazuli CBE throughput encoded in the noise tables may carry an export classification; the scaled JWST numbers stay internal until that is cleared.

Implications. The noise model's normalization is now trustworthy and its photon floor is consistent with what JWST achieves per unit collecting area. The remaining errors are in the inputs and in presentation: an M-dwarf stellar grid that is wrong by tens of percent in the blue, a registry with two inconsistent durations and percent-level ambiguity in k, a default "clear" scenario that no measured planet reaches, and no stellar-contamination term for the hosts where it dominates. None of these is a code bug in the sense of the 2026-10-05 review; all of them change the numbers a user would quote.

Recommendations.

1. Fix the PHOENIX mini-grid (#16) and, for the showcase M dwarfs, normalize the star to Gaia XP over 0.40–1.02 μm rather than to J alone.
2. Make "cloudy" or "high metallicity" the default showcase scenario for sub-Neptunes and label "clear" as an upper envelope in the UI and docs; remove the H_2/He scenarios for TRAPPIST-1 e and LHS 1140 b or show them only with the check-D verdict attached (#8).
3. Resolve the two-path convention (#3) before any public number; the bracketing statistics above differ between paths for that reason alone.
4. Freeze one solution per target with references and uncertainties (#7); HAT-P-11 b and LHS 1140 b first.
5. Add a user-settable stellar-contamination term, defaulting to the check-G bound for late-M hosts, and a systematics floor informed by the JWST achieved-to-predicted ratios (#8).
6. Run checks E, F and H when fitted LD coefficients, H4RG lab data and slicersim time series become available; wire `memo/scripts/` into CI so the tables regenerate with every table change.

Appendix

Reproduction. From the repository root in the `py313` environment:

- `python memo/scripts/fetch_data.py` (network): archive spectra and solutions, SIMBAD photometry, Gaia XP, TRAPPIST-1 e Zenodo files → `memo/data/`.
- `python memo/scripts/checkA_sed.py`, `checkB_photon.py` (needs slicersim), `checkC_geometry.py` (needs batman), `checkD_spectra.py`, `checkG_tlse.py` → `memo/results/*.csv`, `memo/figures/`.
- `python memo/scripts/make_tables.py` → `memo/results/table_*.tex`, which this document inputs.
- The v0.1 pilot scripts (`vegacheck.py`, `combined.py`, `satcheck.py`, `sedcheck.py`, `physcheck.py`, `lsfcheck.py`, `transitcheck.py`) were ported to the `lazuli_exo` name and are superseded by the check scripts above.

Data sources (accessed 2026-10-05).

- NASA Exoplanet Archive, Transmission Spectroscopy table (`transitspec`) and Planetary Systems table (`ps`), TAP service, https://exoplanetarchive.ipac.caltech.edu. The per-dataset reference strings in Tables B2, D1 and D2 are the archive's own; depths are in percent there and converted here.
- Gaia DR3 XP_SAMPLED spectra via the ESA Gaia archive (astroquery); SIMBAD photometry via astroquery.
- Espinoza et al. (2025) data and software, Zenodo record 16125662, https://zenodo.org/records/16125662 (files `appendix-figure1/spectra/JWST_PRISM_Espinoza*.txt`, `HST_WFC3IR_visit*.txt`).

- slicersim 1.0.0 (`0947b53`, 2026-09-17), configuration `lazuli_cbe.toml`: $A = 7.07\,\mathrm{m}^2$, $\eta(\lambda)$, QE(λ), $t_{\mathrm{frame}} = 2.83\,\mathrm{s}$.

References: data used in checks D and G (verified against arXiv abstracts on 2026-10-05).

- Cadieux, C., Doyon, R., MacDonald, R. J., et al. 2024, ApJL 970, L2, "Transmission Spectroscopy of the Habitable Zone Exoplanet LHS 1140 b with JWST/NIRISS", https://arxiv.org/abs/2406.15136. Source of the faculae result quoted in check G.
- Espinoza, N., Allen, N. H., Glidden, A., et al. 2025, ApJL 990, L52, "JWST-TST DREAMS: NIRSpec/PRISM Transmission Spectroscopy of the Habitable Zone Planet TRAPPIST-1 e", https://arxiv.org/abs/2509.05414. Source of the TRAPPIST-1 e spectra in checks D and G.
- Feinstein, A. D., Radica, M., Welbanks, L., et al. 2023, Nature, "Early Release Science of the exoplanet WASP-39b with JWST NIRISS", https://arxiv.org/abs/2211.10493. NIRISS/SOSS spectrum of WASP-39 b (archive rows "Feinstein et al. 2022").
- Lim, O., Benneke, B., Doyon, R., et al. 2023, ApJL, "Atmospheric Reconnaissance of TRAPPIST-1 b with JWST/NIRISS: Evidence for Strong Stellar Contamination in the Transmission Spectra", https://arxiv.org/abs/2309.07047. Source of the 89 ppm precision and the order-of-magnitude stellar-model uncertainty.
- Rustamkulov, Z., Sing, D. K., Mukherjee, S., et al. 2023, Nature, "Early Release Science of the exoplanet WASP-39b with JWST NIRSpec PRISM", https://arxiv.org/abs/2211.10487. PRISM spectrum of WASP-39 b (archive rows "Rustamkulov et al. 2022").
- Sing, D. K., Fortney, J. J., Nikolov, N., et al. 2016, Nature, "A continuum from clear to cloudy hot-Jupiter exoplanets without primordial water depletion", https://arxiv.org/abs/1512.04341. STIS and WFC3 spectra of HD 189733 b, HD 209458 b, WASP-39 b.
- Radica, M., Welbanks, L., Espinoza, N., et al. 2023, MNRAS, "Awesome SOSS: Transmission Spectroscopy of WASP-96b with NIRISS/SOSS", https://arxiv.org/abs/2305.17001. Not used: the spectrum is not in the archive and no public data file was located; WASP-96 b is checked with Nikolov et al. (2018) FORS2 data from the archive.
- Benneke, B., Roy, P.-A., Coulombe, L.-P., et al. 2024, "JWST Reveals CH_4, CO_2, and H_2O in a Metal-rich Miscible Atmosphere on a Two-Earth-Radius Exoplanet", https://arxiv.org/abs/2403.03325. Not used: the TOI-270 d NIRISS spectrum is not in the archive and no public data file was located.
- Other archive datasets in Tables B2, D1 and D2 (Kreidberg et al. 2014, Deming et al. 2013, McCullough et al. 2014, Fraine et al. 2014, Benneke et al. 2019, Tsiaras et al. 2019, Spake et al. 2018, Stevenson et al. 2014, Evans et al. 2016, Wakeford et al. 2018, Nikolov et al. 2016, 2018, and others) are cited as the NASA Exoplanet Archive records them; they were not individually re-verified against the papers.

References: methods and tools (verified).

- De Angeli, F., Weiler, M., Montegriffo, P., et al. 2023, A&A, "Gaia Data Release 3: Processing and validation of BP/RP low-resolution spectral data", https://arxiv.org/abs/2206.06143.
- Montegriffo, P., De Angeli, F., Andrae, R., et al. 2023, A&A, "Gaia Data Release 3: External calibration of BP/RP low-resolution spectroscopic data", https://arxiv.org/abs/2206.06205.
- Rauscher, B. J., Fox, O., Ferruit, P., et al. 2007, PASP, "Detectors for the James Webb Space Telescope Near-Infrared Spectrograph I: Readout Mode, Noise Model, and Calibration Considerations", https://arxiv.org/abs/0706.2344. slicersim's up-the-ramp variance model (Table B1, last column).
- JWST User Documentation, "JWST Telescope", https://jwst-docs.stsci.edu/jwst-observatory-hardware/jwst-telescope: $25.4\,\mathrm{m}^2$ collecting area.
- Kreidberg, L. 2015, PASP, batman (not re-verified in this draft); Winn, J. N. 2010, transit duration and impact-parameter expressions (not re-verified in this draft).

Internal, unpublished

- `2026-09-26_audit_report.md` and `report/2026-10-05_report.md`: the audit and review this memo follows up.
- GitHub issues #3, #7, #8, #12, #16 at `schmidt-observatories/lazuli-exo` (private).